

of the error term for each stock is  $\sigma_{ei}^2$ . Properties (1) and (2) are direct byproducts of a properly specified regression model. Property (3) states that the variance of each factor is  $\sigma_{fk}^2$  and is true by definition. Property (4) states that the error term (residual) and each factor are independent. Analysts need to test to ensure this property is satisfied. Property (5) states that the explanatory factors are independent. Analysts need to properly select factors that are independent or make adjustments to ensure that they are independent. If the factors are not truly independent the sensitivities to these factors will be suspect. Property (6) states that the error terms are independent for all lagged time periods, i.e., no serial correlation or correlation of any lags across the error terms. Property (7) states that the error terms across all stocks are independent, i.e., the series of all error terms are uncorrelated. Since the error term in a factor model indicates company specific returns or noise that is not due to any particular market force these terms need to be independent across companies. If there are stocks with statistically significant correlated error terms then it is likely that there is some market force or some other explanatory variable that is driving returns that we have not accounted for in the model. In this case, while the sensitivities to the selected variables may be correct, some of our risk calculations may be suspect because we have not fully identified all sources of risk. For example, company specific risk, covariance and correlation, and portfolio risk may be suspect due to an incomplete model and may provide incorrect correlation calculations.

When constructing factor models analysts need to test and ensure that all properties are satisfied.

### Matrix Notation

In matrix notation our single stock factor model is:

$$\text{where,} \quad r_i = \alpha_i + Fb_i + e_i \quad (6.33)$$

$$r_i = \begin{bmatrix} r_{i1} \\ r_{i2} \\ \vdots \\ r_{in} \end{bmatrix}, \quad \alpha_i = \begin{bmatrix} \alpha_{i1} \\ \alpha_{i2} \\ \vdots \\ \alpha_{in} \end{bmatrix}, \quad F = \begin{bmatrix} f_{11} & f_{21} & \cdots & f_{k1} \\ f_{12} & f_{22} & \cdots & f_{k2} \\ \vdots & \vdots & \ddots & \vdots \\ f_{1n} & f_{2n} & \cdots & f_{kn} \end{bmatrix}, \quad b_k = \begin{bmatrix} b_{i1} \\ b_{i2} \\ \vdots \\ b_{ik} \end{bmatrix}, \quad e_i = \begin{bmatrix} e_{i1} \\ e_{i2} \\ \vdots \\ e_{in} \end{bmatrix}$$

$r_i$  = vector of stock returns for stock  $i$

$r_{it}$  = return of stock  $i$  in period  $t$

$\alpha_i$  = vector of the constant terms

$F$  = column matrix of factor returns

$f_{jt}$  = factor  $j$  in period  $t$